

Assembly & Machine Language Lab

Assignment 4

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1. Write an Assembly Language Program to add 3 X 3 matrices. Assume the matrices are stored in the form of lists (row wise). First matrix is stored from DS:0030H and the second matrix is stored from DS:0040. Store the result of the addition in the third lists starting from DS:0050H.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov di,0040h
mov bx,0050h
mov cx,0009h
l1:  mov al,[si]
      inc si
      add al,[di]
      inc di
      mov [bx],al
      inc bx
      loop l1
int 03h
mov ah, 4ch
int 21h
main endp
end main
```

Output:

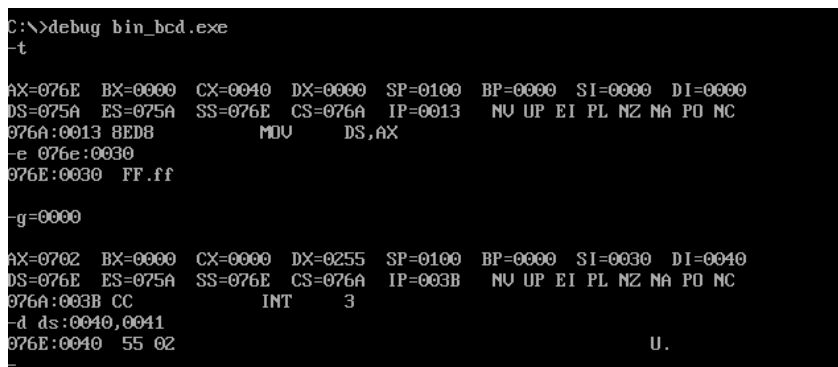
```
C:\>debug add_mat.exe
-t
AX=076D BX=0000 CX=0031 DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076E CS=076A IP=0013 NU UP EI PL NZ NA PO NC
076A:0013 8ED8      MOV     DS,AX
-e 076d:0030
076D:0030 07.1    40.2    50.3    8B.4    C3.5    8C.6    C2.7    05.8
076D:003B 0C.9
-e 076d:0040
076D:0040 B0.1    13.2    50.3    8D.4    86.5    FA.6    FE.7    50.8
076D:004B E8.9
-g=0000
AX=0712 BX=0059 CX=0000 DX=0000 SP=0100 BP=0000 SI=0039 DI=0049
DS=076D ES=075A SS=076E CS=076A IP=002C NU UP EI PL NZ NA PE NC
076A:002C CC      INT     3
-d ds:0050,0058
076D:0050 02 04 06 08 0A 0C 0E 10-12
-

```

2. Write an Assembly Language Program to convert an eight bit binary number stored in DS:0030H into its equivalent BCD number. Stored the result in DS:0040H.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov di,0040h
mov dx,0000h
mov bl,[si]
mov cx,bl
12:  cmp bl,00h
    jz 11
    dec bl
    mov al,dl
    add al,01h
    daa
    mov dl,al
    mov al,dh
    adc al,00h
    daa
    mov dh,al
    loop 12
11:  mov [di],dx
int 03h
mov ah, 4ch
int 21h
main endp
end main
```

Output:



```
C:\>debug bin_bcd.exe
-t

AX=076E BX=0000 CX=0040 DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076E CS=076A IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8ED8      MOV     DS,AX
-e 076e:0030
076E:0030 FF.ff

-g=0000

AX=0702 BX=0000 CX=0000 DX=0255 SP=0100 BP=0000 SI=0030 DI=0040
DS=076E ES=075A SS=076E CS=076A IP=003B  NU UP EI PL NZ NA PO NC
076A:003B CC      INT     3
-d ds:0040,0041
076E:0040 55 02      U.
-
```

3. Write an Assembly program to convert a BCD number stored in DS:0030H into its equivalent hexadecimal number. Stored the result in DS:0040H.

```
dosseg
.model small
.stack 100h
.data
.code
```

```

main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov di,0040h
mov al,[si]
mov bl,al
and al,0f0h
mov cl,04h
ror al,cl
mov dl,0ah
mul dl
mov dx,ax
mov al,bl
and al,0fh
mov ah,00h
add ax,dx
mov [di],ax
int 03h
mov ah, 4ch
int 21h
main endp
end main

```

Output:

```

C:\>debug bcd_hex.exe
-t
AX=076D BX=0000 CX=003A DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076E CS=076A IP=0013 NU UP EI PL NZ NA PO NC
076A:0013 8ED8 MOV DS,AX
-e 076d:0030
076D:0030 01.27

-g=0000
AX=001B BX=0027 CX=0004 DX=0014 SP=0100 BP=0000 SI=0030 DI=0040
DS=076D ES=075A SS=076E CS=076A IP=0035 NU UP EI PL NZ NA PE NC
076A:0035 CC INT 3
-d ds:0040,0040
076D:0040 1B

```

4. Write an Assembly program to convert a binary number stored in DS:0030H into its equivalent gray code. Stored the result in DS:0040H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov di,0040h
mov al,[si]
mov dl,[si]
clc
rcr al,01
xor al,dl
mov [di],al

```

```

int 03h
mov ah, 4ch
int 21h
main endp
end main

```

Output:

```

C:\>debug bin_gray.exe
-t
AX=076C BX=0000 CX=002B DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076D CS=076A IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8ED8      MOV     DS,AX
-e 076c:0030
076C:0030 3D.89
-g=0000
AX=07CD BX=0000 CX=002B DX=0089 SP=0100 BP=0000 SI=0030 DI=0040
DS=076C ES=075A SS=076D CS=076A IP=0026  NU UP EI NG NZ NA PO NC
076A:0026 CC      INT     3
-d ds:0040,0040
076C:0040 CD

```

5. Write an Assembly program to find the factorial of a number stored in DS:0030H. Store the result in DS:0040H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov di,0040h
mov bx,0000h
mov ax,0000h
mov al,[si]
mov cx,[si]
mov bl,al
l1:
    dec bl
    cmp bl,00h
    jz l2
    mul bx
    mov dx,ax
    loop l1
l2:  mov [di],dx
int 03h
mov ah, 4ch
int 21h
main endp
end main

```

Output:

```
C:\>debug FAC.EXE
-t

AX=076D  BX=0000  CX=003B  DX=0000  SP=0100  BP=0000  SI=0000  DI=0000
DS=075A  ES=075A  SS=076E  CS=076A  IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8ED8          MOV     DS,AX
-e 076d:0030
076D:0030  07.07

-g=0000

AX=13B0  BX=0000  CX=4001  DX=13B0  SP=0100  BP=0000  SI=0030  DI=0040
DS=076D  ES=075A  SS=076E  CS=076A  IP=0036  NU UP EI PL ZR NA PE NC
076A:0036 CC          INT     3
-d ds:0040,0041
076D:0040  B0 13          ..
-
```